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Antoine Bonnet

Computer-use agents end to end: the harness, evals and environments behind H's Holo models, and the API and products that ship them.

Experience

Member of Technical Staff, Agents 04/2025 – present
H Company, frontier computer-use agents lab *Paris, France*

- **Agent harness.** Created and lead *SAgent* (60k lines of Python, 59% of commits among 56 contributors), the single agent runtime behind Holo model evals and data, the Agents API, products and forward-deployed work. #1 contributor to H's monorepo: 1.4k merged PRs among 94 engineers
- **Holo4 (co-author).** Rebuilt the harness from OSWorld 2.0 failure analysis: memory compaction for hundred-step tasks (fixed 18 of 19 memory failures), a code + GUI hybrid agent, open-source desktop, browser and *cua* driver CLIs. Holo4: 61.7% on OSWorld 2.0 at 1/7 the cost of Opus 5.5
- **Evals, environments, data.** Integrated 20+ benchmarks (OSWorld 2.0, CUA-Gym, HLE, GDPval, BrowseComp) with nightly gates, LLM-judge failure analysis, RL step rewards and integrity audits. Built trace-to-SFT conversion for any chat format; Holo3.5 prompt ablations gave +7 to +14 pts
- **Surfer 2 and Holo2.** Core contributor to Surfer 2 (orchestrator, judges): state of the art on OSWorld 60.1%, WebVoyager 97.1%, WebArena 69.6%, AndroidWorld 87.1%. Built Holo2's agentic localization: 78.5% on ScreenSpot-Pro (state of the art)
- **Agents API (lead engineer).** Shipped H's public computer-use API 10 weeks after the first PR: 72-operation REST surface (sessions, streaming, webhooks, custom tools), Python and TypeScript SDKs, CLI, MCP server, developer docs
- **Products.** Created HoloTab, a Chrome browser agent (6k users, 4.9/5). Created HoloBricks/HoloBlocks, agents designing LEGO and Minecraft builds up to 27k bricks, shipped in 2 weeks with 11 people. Seeded the open-source HoloDesktop CLI and trajectories.hcompany.ai

Research Scientist Intern 09/2024 – 03/2025
Meta (GenAI) *Paris, France*

- Built a hierarchical multi-agent system with adaptive planning and explicit inter-agent channels; reasoning models beat conventional agents with half the tool calls
- Fine-tuned Llama 3.1 70B by rejection sampling on successful agent trajectories: +6.5% over the post-trained baseline

Data Lead, Meditron 06/2023 – 09/2024
MLO & NLP Labs, EPFL *Lausanne, Switzerland*

- Led the data pipeline for *MEDITRON-70B*: pretraining corpus and open *Clinical Guidelines* dataset; 2.2k GitHub stars, featured on the Meta AI blog
- Built *MOOVE*, a clinician feedback platform for medical LLMs: 1,237 annotators, 22 organizations, 18k contributions

Publications and Open Source

- *Holo4*, H Company, 2026 (co-author); also Holotron-12B, Holo2-235B-A22B
- *Surfer 2: Next Generation of Cross-Platform Computer Use Agents*, Andreux, **Bonnet** et al., 2025
- *Surfer-H Meets Holo1: Cost-Efficient Web Agent Powered by Open Weights*, Andreux, **Bonnet** et al., 2025
- *MEDITRON-70B: Scaling Medical Pretraining for LLMs*, Chen, ..., **Bonnet*** et al., 2023 (*equal contrib.; 850+ citations)
- Creator of hai-agents-python, hai-agents-ts, holo-desktop-cli; merged X11 fixes to trycua/cua (30k stars)

Education

MSc Computer Science, EPFL 09/2022 – 03/2025
Thesis: Multi-Agent Orchestration via Hierarchical Communication, at Meta GenAI *GPA 5.47/6.00*

BSc Joint Honours Math and Computer Science, McGill 09/2019 – 05/2022
First-Class Honours, GPA 3.92/4.00, NSERC Undergraduate Research Award *Montréal, Canada*

Skills

Agents and ML: tool calling, multi-agent systems, LLM judges, RL environments, SFT data, PyTorch, vLLM, Hugging Face
Engineering: Python, TypeScript, Kubernetes, AWS, Docker, REST/OpenAPI, MCP, Chrome extensions